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Enhancing Diversity and Novelty in Book Recommendations: A Multi-Stage Re-ranking Pipeline for Collaborative Filtering

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Declaration of original work

This declaration is made on August 31, 2025.

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Abstract

Recommender systems, which prioritize accuracy in order to help readers navigate an overwhelming array of possibilities, frequently serve to reinforce preexisting preferences. Due to "filter bubbles" and popularity bias, chances for chance discovery are reduced when well-known titles predominate in suggestions. To overcome these constraints in book recommendation, this study creates and assesses a re-ranking pipeline that specifically strikes a balance between accuracy, diversity, and novelty.

Using the Goodbooks-10k dataset, we first establish a baseline Item-Based Collaborative Filtering (IBCF) model. We then apply a multi-stage re-ranking process that integrates three complementary strategies: (1) Maximal Marginal Relevance (MMR) to reduce redundancy, (2) a popularity-aware adjustment to mitigate the dominance of bestsellers, and (3) Explicit Query Aspect Diversification (X-QuAD) using user-generated tags to broaden thematic coverage. Evaluation uses metrics for accuracy (Precision@k, Recall@k), diversity (Intra-List Similarity), and novelty (Catalog Coverage).

Results demonstrate that the pipeline produces significantly more diverse and novel recommendation lists than the baseline IBCF model, with a slight and acceptable trade-off in accuracy. This work highlights the value of post-processing techniques and provides a repeatable framework for creating more human-centered recommender systems that balance accuracy with discovery, ultimately fostering greater user engagement and cultural diversity.

Contents

Chapter 1

Introduction

1.1 Motivation: Beyond the Bestseller List

In the digital age, the wealth of content available offers both enormous potential and a significant obstacle. From modern fantasy novels to philosophical classics, readers can now access millions of volumes from centuries of literary history through online resources like Goodreads, Amazon, and digital library systems. Recommender systems are useful in this situation. By choosing literature based on our personal tastes, they assert themselves as informed advisors in a potentially frightening setting. A well-considered recommendation can change lives by exposing a reader to a new writer, a genre, or an unexpected topic that strikes a deep chord. A reliable bookstore or library who is aware of your tastes and promotes experimentation is frequently imitated by these apps. The best parts of reading are usually these discoveries, which broaden our horizons and improve our ability to communicate across cultural boundaries. Nevertheless, these systems' underlying algorithms have drawbacks.

Most commercial recommendation systems are made with the sole goal of predicting a user's next likely interaction, such as a purchase or high rating. This method typically produces repetitious, limited recommendations, despite its apparent rationality. Orwell's *1984* or Atwood's *The Handmaid's Tale* are two examples of literary classics with comparable themes that a reader who appreciates a particular genre may be exposed to on a regular basis only to the most popular novels in that genre. The "filter bubble," as it is commonly known, confines readers within their own choices by only recommending well-known or popular publications.

The repercussions of this are serious. At the personal level, readers lose out on chances for personal development and surprise. While the "long tail" of equally valuable but less well-known works languishes in obscurity, a tiny number of major hits are recommended everywhere due to algorithmic popularity bias. In literary terms, this can imply that independent writers and voices from marginalized or unusual genres are often ignored. Many recommender systems unintentionally serve as amplifiers of sameness rather than creators of new information. Therefore, the primary goal of this dissertation is not just to improve the accuracy of book recommendations, but also to make them more engaging, varied, and human-centered. In addition to being a prediction engine, a good book recommender should be an informed bookseller who understands your interests and pushes you to try new things.

1.2 The Challenge: Balancing Accuracy, Diversity, and Novelty

A delicate tension between conflicting aims is at the core of recommender system design. This dissertation emphasizes three: accuracy, diversity, and novelty.

Accuracy (Relevance): Recommendations should match the user's preexisting preferences. For instance, a reader of classic science fiction expects to see books like Asimov's *Foundation*, not contemporary romances. The foundation of user trust is accuracy; without it, suggestions are promptly written off as unimportant.

Diversity: Accuracy alone is not enough. Even while a list of nearly identical books may be theoretically correct, it is uninteresting. True diversity requires varied genres, subjects, and viewpoints. For instance, a diverse list might include a classic work of speculative fiction, a modern dystopian novel, and a work combining science fiction and philosophy, rather than five comparable space operas. This variation keeps the user interested and reflects the depth of literary culture.

Novelty: Beyond relevancy and diversity, novelty is the inclusion of fresh, untested material that extends the reader's horizons. Novelty is linked to serendipity, which is the delight of discovering something unexpected but relevant. This means presenting less-known but still important publications to the user. For instance, instead of continuously recommending George R.R. Martin's *A Game of Thrones*, a system might recommend N.K. Jemisin's *The Fifth Season*, a highly acclaimed but lower-selling book.

Finding a balance between these goals is not simple. While algorithms like Item-Based Collaborative Filtering (IBCF) are excellent at determining relevance, they frequently fall short in fostering diversity or novelty. Popularity bias is further reinforced by their heavy reliance on user-item interaction data. This dissertation tackles this problem head-on by investigating how post-processing re-ranking techniques might rebalance suggestions to be not only accurate but also varied and unique.

1.3 Aims and Objectives

The main objective of this project is to develop, implement, and assess a multi-phase recommendation pipeline that improves the novelty and diversity of book suggestions produced by a baseline IBCF model.

To accomplish this goal, four specific objectives are established:

1. **Implementation of the Baseline Model:** To establish a performance benchmark, build a conventional IBCF recommender using the

Goodbooks-10k dataset.

2. **Feature Engineering for Diversification:** Use feature engineering on book tags to construct "aspects"—structured theme representations that serve as the basis for aspect-based diversification.

3. **Design and Integration of a Multi-Stage Re-Ranking Pipeline:** Create a multi-phase re-ranking architecture that combines three tactics:

   - **MMR (Maximal Marginal Relevance):** to strike a compromise between diversity and accuracy.
   - **Popularity-Aware Modification:** to highlight undiscovered treasures and challenge the dominance of bestsellers.
   - **Aspect-Based Diversification (X-QuAD):** to ensure a wide range of topics across themes and genres.

4. **Comprehensive Evaluation:** A thorough comparison between the re-ranked pipeline's performance and the baseline IBCF model will be conducted. A variety of metrics will be used in this assessment to provide a comprehensive picture of the system's performance, including diversity (Intra-List Similarity), novelty (Catalog Coverage), and accuracy (Precision, Recall).

These goals work together to offer a roadmap for developing a recommender system that surpasses the drawbacks of conventional methods.

Chapter 2

Literature Review

2.1 The Landscape of Recommender Systems

In order to assist consumers in navigating vast collections of digital content, recommender systems are an essential subclass of information filtering systems. Usually presented as estimating a rating, ranking, or likelihood of interaction, their primary responsibility is to forecast the level of preference or interest a user would show toward an item. In fields including e-commerce (Amazon, eBay), media streaming (Netflix, Spotify, YouTube), social networking (Facebook, Instagram, TikTok), and news distribution (Google News, Feedly), these systems have been essential over the last 20 years. In order to boost user engagement, decrease cognitive overload, and personalize content distribution, almost all of the main digital platforms in use today use recommendation algorithms. Content-Based Filtering (CBF), Collaborative Filtering (CF), and Hybrid Approaches are the three main categories into which the development of recommender systems has led to the development of a broad variety of approaches.

2.1.1 Content-Based Filtering

Content-based filtering (CBF) compares an item's features to a user's past preferences. Metadata like author, genre, or review keywords can be utilized to create a user profile in a book recommendation system. For instance, the system would suggest more dystopian works with comparable metadata if a reader had expressed interest in dystopian fiction. Because it represents intrinsic item qualities, CBF has a novelty capacity, allowing it to propose books the community has never reviewed, thus solving the cold-start item problem.

On the other hand, CBF is often over-specialized, confining consumers to their preexisting interests. Furthermore, creating and extracting useful item attributes often requires advanced natural language processing or in-depth subject knowledge. Readers may find that information for works misses more nuanced themes or literary styles.

2.1.2 Collaborative Filtering

Collaborative filtering (CF) circumvents the drawbacks of content analysis by utilizing collective wisdom. CF is based on the homophily principle: consumers who have exhibited similar past behavior are likely to appreciate comparable future items. To find latent patterns of taste, CF leverages user-item interaction data (ratings, clicks, purchases) without requiring specific product knowledge. This approach is now the dominant paradigm in academia and business, having shown impressive effectiveness.

2.1.3 Hybrid Approaches

Hybrid systems combine the advantages of both CBF and CF, frequently alleviating each other's shortcomings. For instance, a hybrid pipeline may employ CF to generate candidate items and then use CBF to refine or rerank them using metadata. Netflix is well known for using hybrid techniques, combining user behavior with rich metadata like actors and genres. Similarly, hybridization can enable a system to consider both collaborative behaviors and semantic book properties when making recommendations.

The pipeline suggested in this dissertation can be regarded as a hybrid approach because it starts with an IBCF-based collaborative filtering baseline and then applies re-ranking based on content-derived “aspects” collected from tags.

2.2 Collaborative Filtering in Depth

CF is based on the user–item interaction matrix, a high-dimensional, sparse matrix where each row represents a user and each column represents an item. The entries indicate how strongly a user has interacted with an item, using either explicit (ratings) or implicit (clicks, purchases) data. This representation gave rise to two families of CF models: memory-based (neighborhood) methods and model-based approaches.

2.2.1 Memory-Based Methods (Neighbourhood Models)

Neighborhood models work directly on the user–item matrix without a separate learning phase.

User-Based Collaborative Filtering (UBCF): This technique forecasts preferences for a target user by averaging the evaluations of “neighbor” users with similar interaction histories. While simple, UBCF is not very scalable; the computational cost of finding neighbors grows with the number of users. UBCF also falters on sparse datasets where user overlaps are inadequate for a trustworthy similarity calculation.

Item-Based Collaborative Filtering (IBCF): Introduced by Sarwar et al. (2001) [12], this focuses on item similarity. The assumption is that two items are likely similar if consumed by many of the same users. Metrics like Pearson Correlation and Cosine Similarity are used to calculate similarity. For example:

$$\text{similarity}(i, j) = \cos(\mathbf{v_i}, \mathbf{v_j}) = \frac{\mathbf{v_i} \cdot \mathbf{v_j}}{\|\mathbf{v_i}\| \cdot \|\mathbf{v_j}\|}$$

where $\mathbf{v_i}$ is the interaction vector of item i.

IBCF is superior to UBCF because item-to-item similarities are more stable over time than user preferences, and recommendation creation is made more efficient by pre-calculating an item-to-item similarity matrix. These factors make IBCF the foundational model in this dissertation.

2.2.2 Model-Based Methods

Model-based CF uses the user–item matrix to learn latent representations. The most well-known are Matrix Factorization (MF) techniques, which gained popularity during the Netflix [7]. A matrix A is approximated by multiplying two lower-rank matrices:

$$A \approx U \cdot V^T$$

where U denotes user factors and V denotes item factors. The user's preference for item j is approximated by the dot product $u_i \cdot v_j$. While MF reflects hidden preference characteristics (e.g., "likes dystopian fiction"), it can reinforce popularity bias since latent variables are learned more firmly for popular items.

Recent developments include neural CF, where deep neural networks replace dot products, and graph-based CF. Despite their ability to capture intricate patterns, these techniques are still susceptible to the beyond-accuracy issues discussed below.

2.3 The Challenge of Beyond-Accuracy Metrics

The goal of recommender research has historically been to maximize accuracy metrics such as Root Mean Squared Error (RMSE), Precision, and Recall. However, systems that are only tuned for accuracy can produce subpar user experiences. Users prefer variety, surprise, and discovery over a list of nearly identical choices. This has increased research on metrics beyond accuracy, such as serendipity, diversity, and novelty.

2.3.1 Popularity Bias and the Long Tail

Popularity bias is a persistent issue. Popular items receive more interactions, increasing their likelihood of being recommended. This creates a feedback loop where popular items get more recommendations, boosting their popularity further, while obscure "long tail" items are rarely seen. In the book world, this leads to countless suggestions for major bestsellers, while thousands of excellent niche books remain undiscovered. [3]

2.3.2 Novelty and Serendipity

To combat popularity bias, researchers emphasize novelty (exposing users to items they haven't seen) and serendipity (making unexpectedly relevant recommendations). Novel suggestions broaden users' perspectives, while serendipitous ones bring joy. Methods for achieving novelty include using diversity constraints, exploring less-traveled graph pathways, and down-weighting popular items. This dissertation's popularity-aware re-ranking stage specifically addresses this problem.

2.4 Re-Ranking for Diversity

Re-ranking is a post-processing technique that separates candidate generation from the final list's optimization. A relevance-focused model, like IBCF, first generates a candidate pool. A re-ranking algorithm then rearranges this list to maximize beyond-accuracy objectives.

2.4.1 Maximal Marginal Relevance (MMR)

MMR is a method for ranking items that balances diversity and relevance. Initially developed for text summarization, MMR helps ensure that chosen items are both relevant to the user's query and distinct from one another, minimizing [2].

The procedure is iterative. At each stage, an item is selected based on a scoring function considering its relevance to the query and its dissimilarity from previously chosen items. A parameter λ manages this trade-off. Formally, the selection is:

$$\text{MMR} = \arg\max_{i \in R \setminus S} \left[\lambda \cdot Rel(i,u) - (1-\lambda) \cdot \max_{j \in S} Sim(i,j) \right]$$

where R is the candidate pool, S is the set of already selected items, $Rel(i,u)$ is the relevance of item i to user u, and $Sim(i,j)$ is the similarity between items. MMR enhances result quality by balancing coverage and precision.

2.4.2 Aspect-Based Diversification (X-QuAD)

X-QuAD (Explicit Query Aspect Diversification) was created for web search to ensure multiple aspects of a query are represented in the results. The core idea is that user queries are often vague, and different users may be interested in different underlying interpretations. X-QuAD addresses this by iteratively selecting items, weighing their overall significance against the extra coverage they offer for underrepresented aspects. The score for a candidate item d is given by [11]:

$$\text{X-QuAD}(d) = (1-\lambda)\cdot P(d \mid q)+\lambda\cdot \sum_{a\in A} P(a \mid q)\, P(d \mid q, a) \prod_{d'\in S} \big(1-P(d' \mid q, a)\big)$$

The trade-off parameter λ balances general relevance and aspect coverage. In recommendation, aspects can be extracted from tags, metadata, or categories. X-QuAD broadens the thematic scope of suggestions by rewarding items that highlight underrepresented elements.

2.4.3 Other Diversification Methods

Researchers have also explored Determinantal Point Processes (DPPs), which favor varied subsets by modeling negative correlations, and submodular optimization, which offers theoretical guarantees for coverage maximization. Additionally, bandit-based methods that frame diversity as an exploration-exploitation trade-off have been investigated.

2.5 The State of the Art in Book Recommendation

Book recommendation has different challenges than recommending movies or music. Books are longer, read less frequently, and have more complicated themes. Book consumption is also often impacted by social factors, with communities like Goodreads being crucial. Early book recommenders heavily used IBCF, as shown by Amazon in the 2000s [9]. Later systems added metadata like authors and genres. Recent work has introduced NLP-based models using transformer-based embeddings (like BERT) or topic modeling (like Latent Dirichlet Allocation) to represent book content semantically. Deep learning techniques like autoencoders and graph neural networks (GNNs) have shown promise but are susceptible to popularity bias, computationally costly, and often opaque. Therefore, re-ranking techniques like MMR and X-QuAD remain useful supplements.

2.6 Identifying the Research Gap

Despite extensive literature, several gaps remain. First, while individual diversification techniques are well-studied, their cumulative sequential effects are not. Few studies assess the interplay between different re-ranking methods used in series. Second, re-ranking is only gradually being adapted from information retrieval to recommendation domains, especially with rich semantic features like tags. Third, evaluation frameworks often focus on a single metric rather than the interaction of novelty, diversity, and accuracy.

This dissertation addresses these gaps by:

- Creating a unique sequential pipeline combining three re-ranking techniques, each targeting a distinct IBCF vulnerability.
- Using the pipeline in the culturally diverse field of book recommendations, where subject diversity is heavily emphasized.
- Performing a thorough analysis using metrics for accuracy, diversity, and novelty to offer a sophisticated understanding of trade-offs.

Chapter 3

Data Description

3.1 Dataset Selection and Justification

The choice of dataset is foundational to any data-driven project. After weighing numerous options, the Goodbooks-10k dataset, compiled from Goodreads, was chosen for this investigation. Over 48,871 users, 9,986 books, and a wealth of user-generated tagging data are all included. It is a good fit for a project that prioritizes suggestion diversity because of its harmony of accessibility, richness, and scale.

3.1.1 Rationale for Dataset Selection

While popular datasets like Amazon Reviews, Last.fm, and MovieLens are common in recommender system studies, they are not ideal for this dissertation's goals. Datasets centered on movies or music capture different consumption habits than those for books, which are read over longer periods and have a wider genre range. Additionally, Amazon's purchase data may not accurately reflect personal interest due to factors like gift-giving, and its product data lacks the deep semantic metadata available for books [10].

In contrast, Goodbooks-10k provides:

- **Topical and stylistic diversity:** The heterogeneity of reading, where consumers frequently consume literature from a variety of genres (such as fantasy, classics, and romance).
- **Rich semantic signals:** The thematic nuance needed for aspect-based variation is provided by user-generated tags (such as *magical-realism* and *dystopian*).
- **Implicit preference data:** "To-read" shelves are powerful predictors of intent, creating strong user-item interactions where formal ratings may be scarce.

3.2 Dataset Composition

The dataset comprises four linked files central to this work:

- **to_read.csv** — The core user-item interaction file, containing `user_id` and `book_id` pairs. A user adding a book to their "to-read" shelf is represented by each row.
- **books.csv** — Contains complete metadata, including title, author, average rating, and year of release, for every one of the 10,000 books.
- **tags.csv** — A dictionary that maps `tag_id` into human-readable `tag_name`.
- **book_tags.csv** — Connects books to tags with a count of how many users applied each tag to a book, enabling aspect extraction.

These files come together to provide a multi-layered dataset that includes user-book interactions, semantic content, and metadata.

3.3 Data Pre-processing and Cleaning

Due to problems with sparsity and duplication in the raw Goodreads data, a systematic pre-processing pipeline is required.

3.3.1 Merging and Deduplication

To create a single master DataFrame, the four source files were combined. Duplicate user-book pairs were eliminated to prevent artificially inflating interaction counts, keeping only the first interaction.

3.3.2 Addressing Sparsity

Exploratory Data Analysis revealed a long-tail distribution where many users interacted with few books, and vice versa. This is an issue, as sparse items lack sufficient interaction history for reliable similarity calculations, and sparse users provide too little information to learn preferences. To address this, thresholds were applied:

- **User filtering:** Users with fewer than 10 interactions were removed.
- **Book filtering:** Books with fewer than 20 interactions were removed.

This filtering balanced dataset density with coverage, ensuring the model trained on active users and reasonably popular books.

3.3.3 Resulting Dataset

After filtering, the dataset became smaller but denser, improving its reliability for collaborative filtering. This filtered dataset was saved as a reproducible CSV to ensure consistency across experiments.

3.4 Aspect Feature Engineering

Feature engineering for diversification was a key step in this project. While books lack explicit topics, the X-QuAD algorithm requires item aspects (themes or subtopics). User-generated tags on Goodreads serve as a powerful proxy.

3.4.1 Process

1. Merge `book_tags.csv` with `tags.csv` to map tags to names.
2. For each book, rank tags by frequency of application.
3. Select the top 10 tags as representative aspects.
   - Fewer tags risk missing thematic nuance.
   - Too many tags introduce noise.
   - Ten tags offered a balanced representation.
4. Store as a new aspects feature in the master DataFrame.

For example, a book like *Going After Cacciato* might yield aspects: *fiction*, *classics*, *historical-fiction*, *war*, *Vietnam*, *historical*, *non-fiction*, *owned*, *literature*, *history*.

This representation enables the X-QuAD re-ranking stage to diversify recommendations across themes, ensuring lists reflect the breadth of a user's interests.

3.5 Exploratory Data Analysis (EDA)

To gain a comprehensive grasp of the underlying structure and properties of the dataset, a comprehensive Exploratory Data Analysis (EDA) was carried out. Important preprocessing decisions, especially the filtering levels used in

Section 3.3, were informed by this study, which was essential for placing the data in perspective.

3.5.1 Interaction Distribution

Findings and Analysis: A clear power-law distribution was shown by the histograms for interactions per book and user-book interactions. A "long tail," which denotes a notable imbalance in activity, characterizes this. Interactions per User: Most of the interactions (e.g., adding books to their "to-read" shelves) are carried out by a few very active users (referred to as "power users"). The lengthy tail of the distribution is caused by the vast majority of users interacting infrequently.

Interactions per Book: Similar to this, a select few well-known books are designated as "to-read" by a large number of people, making them the network's key nodes. There aren't many interactions with most other books.

This skewed distribution demonstrates the data's sparsity and supports the necessity of filtration. We can eliminate the least active people and obscure books by establishing minimum interaction thresholds (e.g., individuals with at least 10 interactions and books with at least 20), which would produce a denser and more trustworthy dataset for developing a recommendation model.

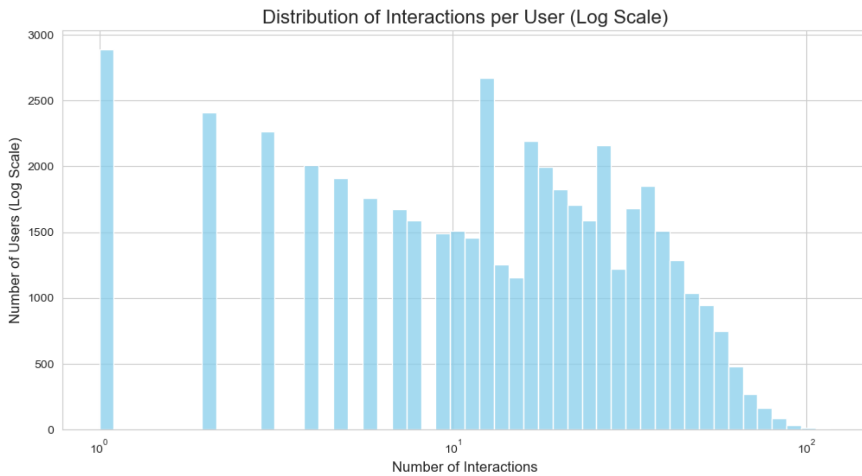

Figure 3.1: Distribution of User Interaction Frequency

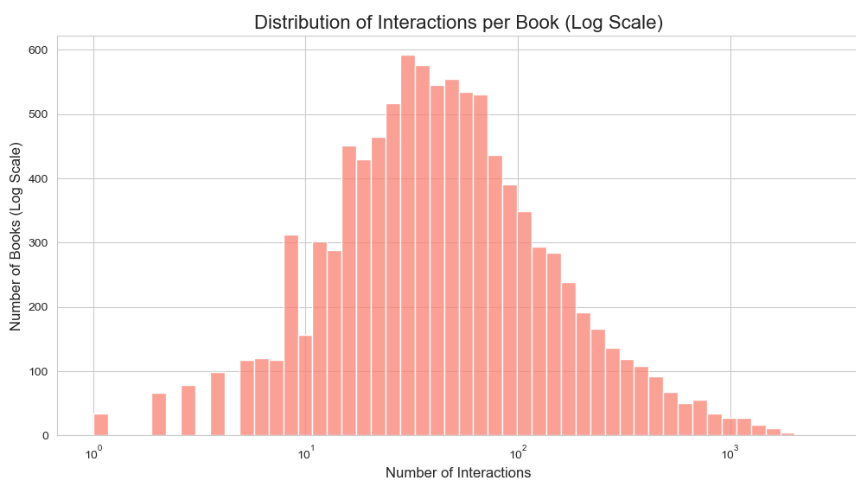

Figure 3.2: Distribution of Book Interaction Frequency

3.5.2 Popular Tags (Genres)

Findings and Analysis: The genre bias in the dataset is evident in the bar chart of the top 20 book tags, which represents the main interests of the Goodreads user base.

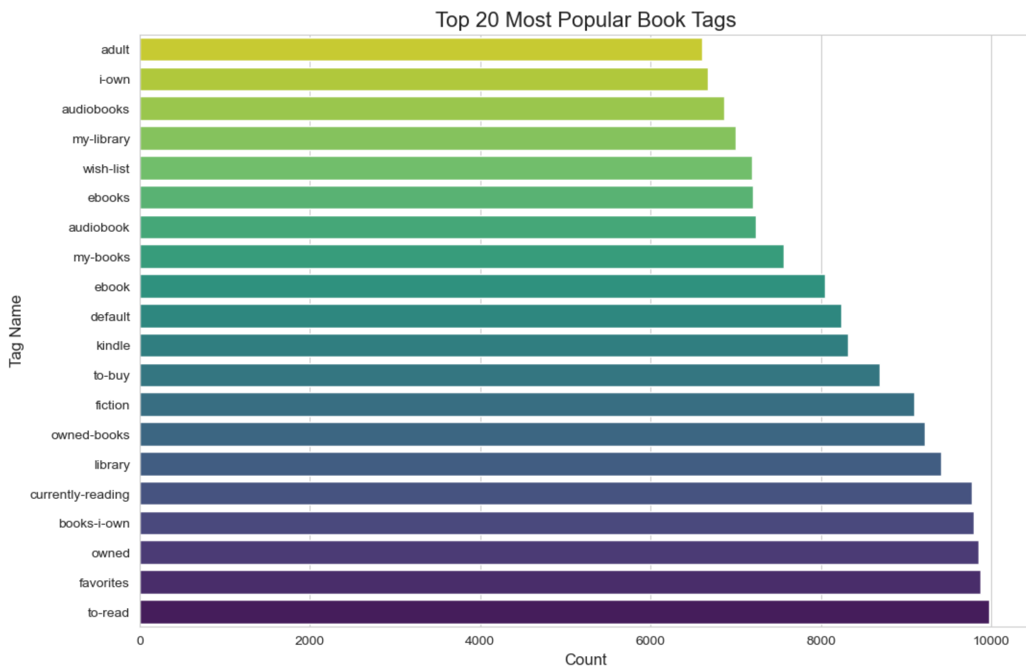

Figure 3.3: Frequency of Top 20 Most Common Book Tags

- Dominant Genres: The most common tags are young adult, fantasy, and fiction, suggesting a strong inclination toward well-liked modern genres. This implies that books in these popular categories will be the ones the model recommends the most.

- Genre Diversity: There is still a significant amount of topic diversity, as evidenced by the inclusion of more specialist classifications like historical fiction and non-fiction in the top 20. This guarantees that the dataset can provide recommendations across a broad variety of interests and is not totally monolithic.

3.5.3 Ratings Distribution

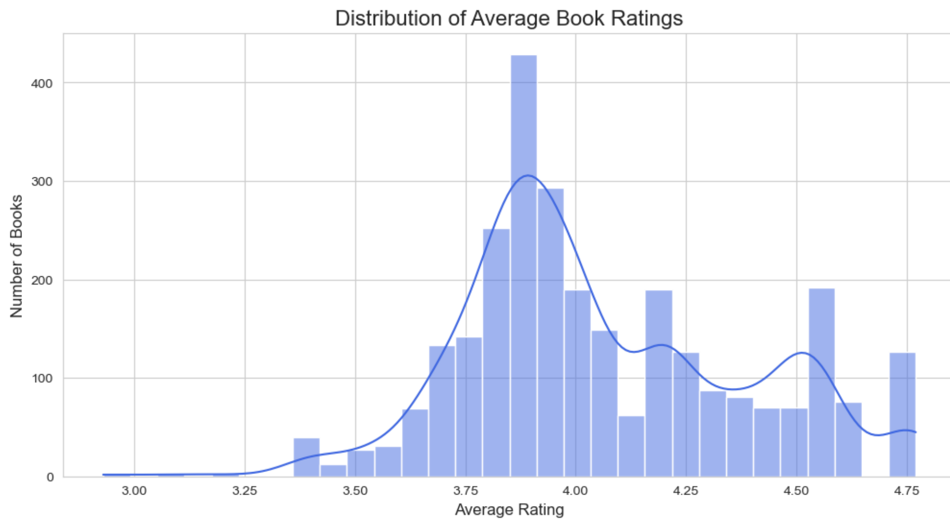

Figure 3.4: Distribution of Average Book Ratings

Findings and Analysis: The distribution of average book ratings is heavily skewed towards higher scores, with most books rated between 3.5 and 4.5. This pattern suggests an 'optimism bias,' common on review platforms, where users who enjoy a book are more likely to rate it. This inflates average scores and indicates that raw ratings may not be sufficient to distinguish between 'good' and 'great' books without normalization.

3.5.4 Publication Year Trends

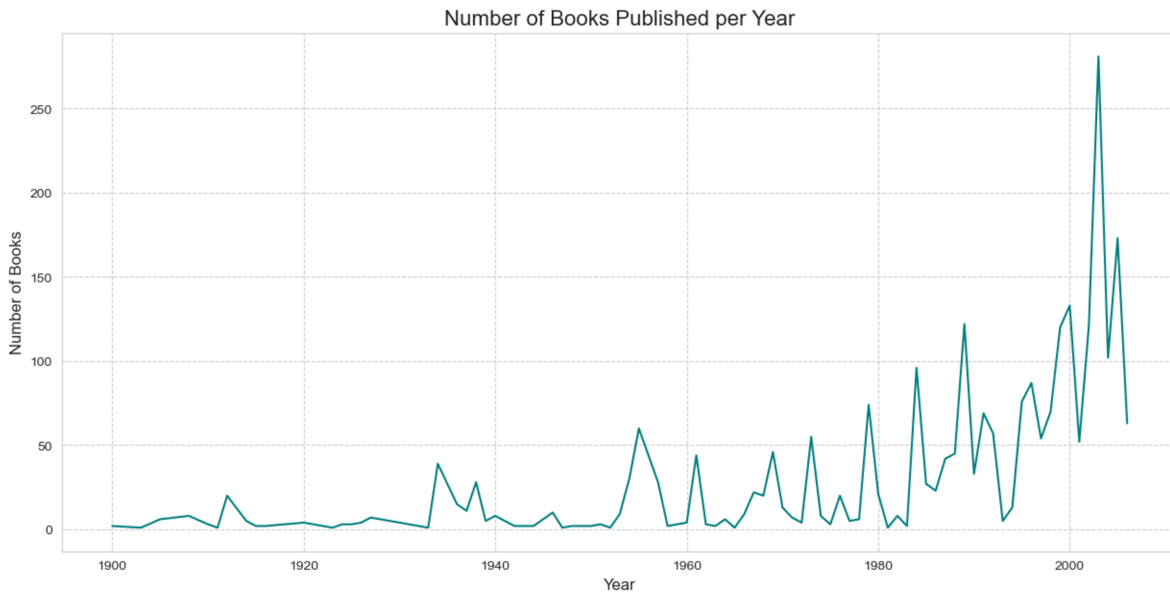

Figure 3.5: Distribution of Books by Original Publication Year

Findings and Analysis: Books published in the 20th and 21st centuries are significantly more represented in the line graph that counts books by the year of original publication.

- **Contemporary Focus**: The majority of the dataset's pieces are contemporary, with a peak in recent decades. Although there is some classic literature from previous centuries, it is significantly less common. This suggests that the recommendation algorithm will be more suited to modern writing than to old works.

3.5.5 Author vs. Genre Heatmap

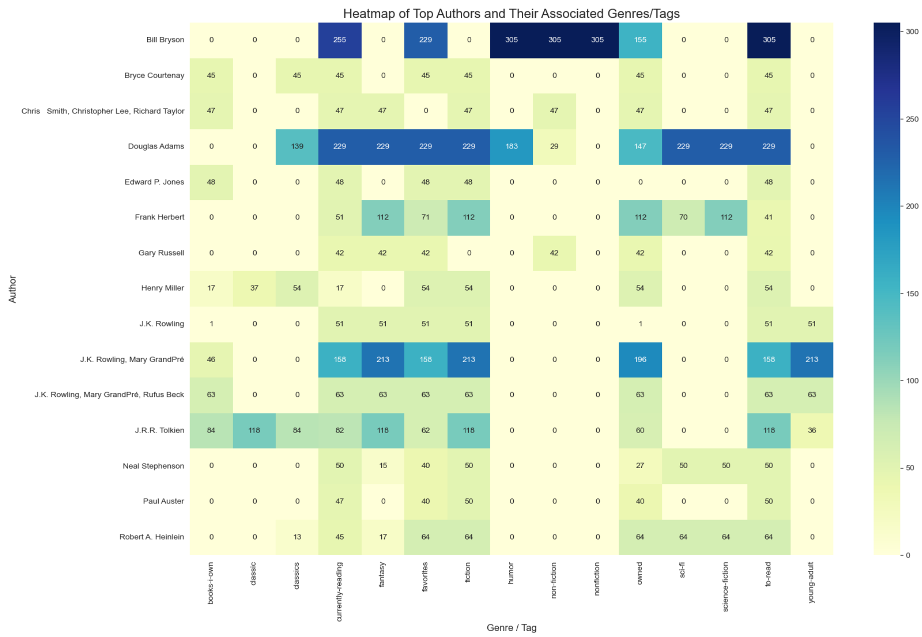

Figure 3.6: Genre Association Heatmap for Top Authors

Findings and Analysis: The author-genre heatmap reveals strong clustering, confirming that popular authors in the dataset often specialize in specific genres. For example, George Orwell is highly associated with 'classics' and 'dystopian,' while J.K. Rowling is linked to 'fantasy' and 'young-adult.' This genre-centric user preference structure provides a crucial signal for the recommendation model.

Chapter 4

Methodology

4.1 System Architecture and Experimental Design

The approach employed in this dissertation is predicated on a comparative analysis of two different recommendation pipelines: the suggested multi-stage re-ranking pipeline intended to improve variety and innovation, and a conventional baseline system that just considers relevance. The purpose of this experimental design was to specifically identify and quantify the incremental advantages of advanced re-ranking techniques. It is feasible to directly link any observed variations in results to the diversification techniques implemented by maintaining the same core collaborative filtering model throughout both pipelines and simply altering the post-processing re-ranking processes. The main idea that a multi-stage strategy can effectively balance the frequently conflicting goals of accuracy, diversity, and novelty in a book recommendation setting is validated by this controlled comparison. The following is a summary of the two pipelines being examined:

1. **Baseline Pipeline ("IBCF Only")**: This system is an example of a traditional and popular recommendation system technique. The items

with the highest similarity ratings to a user's seed book are chosen to create a top-N suggestion list. The collective knowledge of user behavior is captured by a pre-computed item–item similarity matrix, from which the similarity scores are obtained. The baseline is therefore a crucial benchmark that is solely concerned with optimizing for relevance (i.e., accuracy). This study aims to improve upon the current state of affairs, especially in areas that go beyond mere resemblance.

2. **Full Pipeline (Multi-Stage Re-Ranking)**: The dissertation's main contribution is embodied in this system. The same IBCF model is used to create the original, larger candidate list. After that, this list goes through three different re-ranking phases, each of which is intended to remedy a particular drawback of conventional recommenders:

    - **Stage 1 (MMR - Maximal Marginal Relevance)**: The redundancy issue is addressed in this first phase. In order to promote intra-list diversity, it rearranges the candidate list to exclude items that are overly similar to one another (such as several books from the same series).
    - **Stage 2 (Popularity Adjustment)**: The tendency of recommenders to over-represent blockbusters is known as popularity bias, and it is addressed at this step. It gently penalizes very popular things, allowing "hidden gems" that are less well-known but nevertheless important to emerge.
    - **Stage 3 (X-QuAD - Explicit Query Aspect Diversification)**: Topical coverage is ensured in the last step. In order to guarantee that the final recommendations cover a range of user interests, it reranks the list using user-generated tags as stand-ins for various topics and genres.

The same preprocessed dataset, the same set of seed items for recommendation generation, and the same evaluation metrics are used to assess both pipelines under the same experimental setup. The study can precisely measure the influence of each diversification method on the end user experience thanks to this meticulous arrangement, which guarantees a fair and direct comparison.

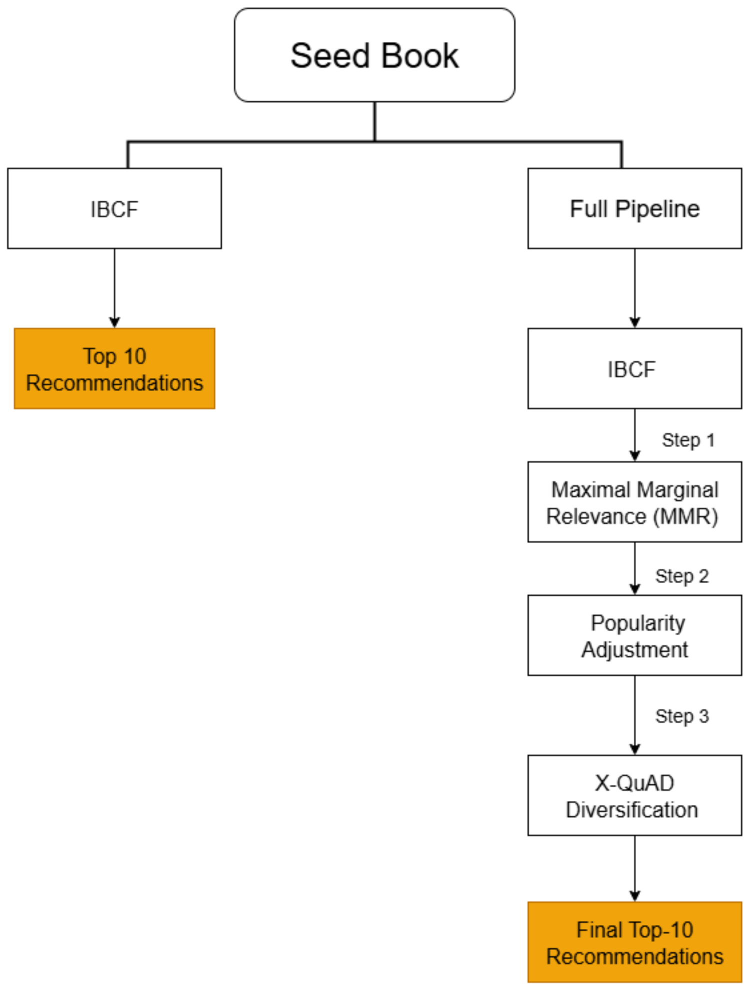

Figure 4.1: Architectural Comparison of the Baseline and Multi-Stage Re-Ranking Pipelines

4.2 Core Component: Item-Based Collaborative Filtering (IBCF)

The foundation for both the baseline and the proposed re-ranking pipelines is an Item-Based Collaborative Filtering (IBCF) model. This algorithm was selected as the core recommendation engine for several key reasons. Firstly, it is a well-established and computationally efficient model, particularly effective in scenarios where the number of items is relatively stable. Secondly, its reliance on an item-item similarity matrix makes it highly interpretable; the reason for a particular recommendation can be easily explained by its similarity to a user's known preferences. This transparency is crucial for understanding and refining the subsequent re-ranking stages. Finally, IBCF's performance as a standalone model provides a strong and relevant benchmark for measuring the added value of the diversification strategies.

4.2.1 Matrix Construction and Data Representation

The user–item interaction matrix, represented by the letter A, is the central component of both pipelines. The unique books in the dataset are represented by the columns of this matrix, and the unique users are represented by the rows. The to_read.csv file, which records a user's intention to read a specific book, is the source of the interactions.

Implicit Signal and Binary Representation The matrix's entries are binary since the interaction signal is implicit (a user placing a book to a "to-read" shelf) as opposed to explicit (a star rating ranging from 1 to 5). The collaborative filtering logic is based on this "to-read" action, which is a powerful gauge of user interest. Thus, the definition of the matrix entries is:

$$Au, i = \begin{cases} 10 & \text{if user } u \text{ marked book } i \text{ as ``to-read''}, \\ 0 & \text{otherwise.} \end{cases}$$

Sparsity and Computational Efficiency This matrix is extremely sparse, as was determined in the Exploratory Data Analysis (Chapter 3). With 48,871 distinct users and 9,986 distinct books, the raw dataset creates a matrix with more than 488 million potential interactions. However, the sparsity of the matrix is about 99.81% because only 912,705 of these interactions are non-zero. There are opportunities as well as challenges associated with this great sparsity. It would be computationally costly to store the entire matrix in memory. As demonstrated in the project's code, the matrix is built as a Compressed Sparse Row (CSR) structure using Python's SciPy package in order to handle this effectively. Only the non-zero values and their accompanying indices are stored in the CSR format, significantly lowering memory needs and speeding up the linear algebra calculations required to determine item similarities. The computation can be performed on normal hardware thanks to this crucial implementation detail.

Data Transformation: From Raw IDs to Matrix Indices The raw user_id and book_id values from the dataset must be converted into a collection of continuous integer indices appropriate for matrix operations before the matrix can be built. The scikit-learn library's LabelEncoder is used to do this. Each distinct user and book ID is mapped to a distinct integer index between 0 and N-1 in this process, where N is the total number of distinct users or books in the filtered training data. This stage guarantees the most compact sparse matrix feasible.

4.2.2 Item–Item Similarity Calculation

The item–item similarity matrix, S, is the central component of the IBCF model. Each entry Si,j indicates how similar two books, i and j, are to one another.

- **Similarity Metric**: Cosine Similarity was chosen since it works well with limited data. It is resilient to variations in item popularity since it calculates the cosine of the angle between two item vectors (vi, vj):

$$\text{similarity}(i, j) = \cos(\mathbf{v_i}, \mathbf{v_j}) = \frac{\mathbf{v_i} \cdot \mathbf{v_j}}{\|\mathbf{v_i}\| \, \|\mathbf{v_j}\|}$$

- **Computation**: The item-user interaction matrix must first be L2-normalized in order to calculate this effectively. Next, a single matrix multiplication is used to determine the similarity matrix S, which is the dot product of the normalized matrix and its transpose:

$$S = A_{\text{norm}} \cdot A_{\text{norm}}^T$$

During the recommendation process, this action produces a dense, symmetric 9,986 x 9,986 matrix that is pre-computed to enable quick retrieval of related items.

4.2.3 Candidate Generation

The procedure of creating recommendations for a specific seed book online is quite effective after the item-item similarity matrix S has been pre-calculated. The process entails obtaining the row in S that corresponds to the seed book, which includes its similarity scores with every other book in the catalog. The novels are then ranked from most to least similar using these ratings, which are subsequently sorted in descending order. The candidate list is created by choosing the top-N entries. A crucial step in this process is removing the seed book from the list of recommendations, as it will always have a perfect similarity score of 1.0.

This candidate generation step, implemented in the `get_ibcf_recommendations` function, forms the complete recommendation logic for the Baseline Pipeline and serves as the initial input for the Full Pipeline's re-ranking stages.

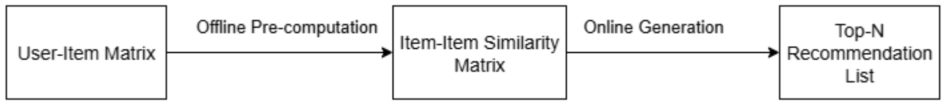

Figure 4.2: Workflow of the Core IBCF Recommendation Model

4.3 Multi-Stage Re-Ranking Pipeline

The initial candidate set of suggestions produced by the IBCF model will be methodically improved and refined using the suggested workflow. To guarantee a wide range of potential diversification candidates, it starts with a bigger pool of 50 pertinent books. Three successive re-ranking steps are then applied to this list, each of which aims to address a distinct drawback of conventional collaborative filtering.

Stage 1: Maximal Marginal Relevance (MMR) for Intra-List Diversity The typical issue of redundancy in recommendation lists is addressed in the first re-ranking step. The usefulness and potential for user discovery are sometimes limited because the top-ranked items from a similarity-based model are frequently quite similar to one another (for example, multiple novels from the same series). To address this, a fresh, diverse list is iteratively created using Maximal Marginal Relevance (MMR) [2]. The original relevance score of an item is weighed against how similar it is to other items that have already been chosen for the list. The MMR equation is:

$$\text{MMR} = \arg\max_{i \in R \setminus S} \left[\lambda \cdot Rel(i, u) - (1 - \lambda) \cdot \max_{j \in S} Sim(i, j) \right]$$

- **Parameter Setting**: The value of λ, the trade-off parameter, was 0.85. This value was selected to provide a meaningful penalty to items that are too similar to those already in the new list (S) while maintain-

Book Title	Original IBCF Score	Normalized Popularity	Adjusted Score	Rank Change
Going After Cacciato	1	0	0.9	No Change
Into the Wild	0.707	1	0.736	No Change
Persepolis	0.707	0.655	0.702	No Change
Harry Potter	0.156	0.899	0.23	Up
J.R.R. Tolkien Box Set	0.183	0.021	0.166	Down

Table 4.1: Comparison of baseline and adjusted scores for selected books.

ing a strong emphasis on the original relevance score (Rel(i,u)).

- **Effect**: After processing the first 50 applicants, the list is narrowed down to the top 20, producing a set of suggestions that are both internally varied and pertinent [8].

Stage 2: Popularity Adjustment for Novelty The purpose of this stage is to counteract popularity bias, which occurs when recommendation systems over-represent internationally well-known blockbusters. A popularity adjustment is made to highlight lesser-known titles and encourage novelty. Each book's ratings_count, which is then normalized to a scale of [0, 1], is used to determine how popular it is. The adjusted score is calculated as [1]:

$$\text{score}_{\text{pop}} = (1 - \alpha) \cdot \text{ibcf\_score} + \alpha \cdot \big(1 - \text{norm\_popularity}\big)$$

- **Parameter Setting**: 0.1 was the value of the parameter α. As a tiebreaker for items with comparable IBCF scores, this tiny value guarantees that the change has a mild corrective effect. Less well-known books are subtly elevated without significantly changing the relevance-based ranking.

- **Effect**: This new score is used to re-sort the 20-item list from the MMR stage, boosting the likelihood that less well-known but pertinent books would rank higher.

Stage 3: X-QuAD Diversification for Topical Coverage The last step ensures that the list of suggested readings encompasses a wide range of subjects and genres. It employs X-QuAD (Explicit Query Aspect Diversification), which makes use of each book's rich, user-generated tags as stand-ins for various "aspects" or topics. The final list is iteratively constructed by the X-QuAD algorithm, which rewards items that add new, underrepresented categories. This is the scoring function:

$$\text{score}_{\text{X-QuAD}} = \lambda \cdot \text{relevance} + (1 - \lambda) \cdot \text{aspect\_gain}$$

where aspect_gain is a measure of how many new tags an item contributes to the set of tags already covered by the list.

- **Parameter Setting**: With the parameter λ set at 0.85, the original relevance score was once more given priority, but items that expanded the suggestions' thematic area were encouraged to be included.
- **Effect**: The final top-10 list, which is intended to be pertinent, non-redundant, and topically diversified, is created by processing the popularity-adjusted list [8].

4.4 Implementation Details

The project was implemented in Python using standard data science libraries: Pandas for data manipulation, NumPy and SciPy for efficient sparse matrix operations (CSR format), and Scikit-learn for preprocessing tasks like LabelEncoder and normalization. Visualizations were generated with Matplotlib and Seaborn.

The implementation was conducted with a rigorous adherence to a modular design. Separate, clearly described functions contained the fundamental reasoning behind the IBCF model as well as the re-ranking techniques (mmr_re_rank, apply_popularity_boost, and xquad_re_rank). This method was essential to the experimental design since it made it possible to construct pipelines in a flexible manner and made it easier to compare the baseline and the complete model. It also made the code easier to read and maintain. By choosing turning on or off particular steps, this modularity also simplifies the process of performing ablation research.

4.5 Evaluation Framework

To measure the two pipelines' performance quantitatively, a strong and comprehensive evaluation framework was developed. The framework was created to capture the crucial trade-offs between innovation, diversity, and relevance in addition to evaluating how accurate the recommendations were. To fully comprehend the effects of the multi-stage re-ranking pipeline, a comprehensive approach is necessary [6].

4.5.1 Train–Test Split

A rigorous user-centric approach was used to divide the dataset into training (80%) and testing (20%) sets in order to guarantee an objective and realistic assessment. Because it avoids data leaks and more closely mimics a real-world situation where a system must suggest new items to a user based on their past behavior, this approach is better than a straightforward random split of encounters. The split was carried out by partitioning the list of "to-read" interactions for every user in the dataset. Twenty percent of their interactions were reserved for the test set, while the remaining eighty percent were assigned to the training set. One item from each test user's held-out collection served as the seed item, from which suggestions were derived. The ground truth, or the collection of pertinent items that a perfect recommendation system should detect, was made up of the remaining items they held out.

4.5.2 Evaluation Metrics

To give a thorough picture of each pipeline's performance in relation to the three main dimensions of accuracy, diversity, and Novelty, a set of metrics was selected. For all measures, a standard recommendation list size of k=10 was used [4].

Accuracy: Precision@k and Recall@k The relevancy of the top-k recommendations is assessed using these common information retrieval metrics [5].

- Precision@k calculates the proportion of pertinent recommendations. "Of the 10 books recommended, how many were actually in the user's ground truth set?" is the question it addresses.

- Recall@k calculates the percentage of pertinent things that make it into the list of recommendations. The reply is: "Of all the books the user would have liked, how many did we successfully recommend in the top 10?"

$$\text{Precision@}k = \frac{|\text{Recommended} \cap \text{Relevant}|}{k}$$

$$\text{Recall@}k = \frac{|\text{Recommended} \cap \text{Relevant}|}{|\text{Relevant}|}$$

Diversity: Intra-List Similarity (ILS@k) Intra-List Similarity (ILS) was computed to assess the suggestion list's variety. The average pairwise cosine similarity between every item in the top-k list is calculated by this metric.

$$\text{ILS@}k = \frac{2}{k(k-1)} \sum_{i \in R_k} \sum_{\substack{j \in R_k \\ j \neq i}} Sim(i,j)$$

Higher variety is indicated by a lower ILS score, which implies that the suggested items cover a wider range of material and are less identical to one another.

Novelty: Catalog Coverage Catalog Coverage was tested to assess the system's capacity to suggest a broad range of products outside of the best-selling goods. The percentage of the entire item catalog that is suggested to at least one user in the test set is determined by this metric.

$$\text{Coverage} = \frac{\left|\bigcup_{u \in U_{\text{test}}} R_u\right|}{|\text{Catalog}|}$$

Greater Novelty is indicated by a higher coverage score, which demonstrates that the system can surface a wider range of the catalog, including less well-liked items.

4.6 Computational Complexity

Each stage's computational properties were examined to make sure the suggested pipeline is both efficient and effective. The best way to understand the complexity is to distinguish between the real-time online expenses and the one-time offline charges. With a complexity of roughly O(|I|2·d), the offline computation of the item-item similarity matrix is the most time-consuming phase. Despite its demands, this pre-computation process is doable given the size of the dataset, particularly when using efficient sparse matrix operations. There aren't many expenses involved in creating a recommendation list online. With efficient complexity of O(k2) for MMR, O(k) for the popularity adjustment, and O(k·m) for X-QuAD, the re-ranking stages work on a tiny candidate set (k). This small overhead guarantees that the pipeline as a whole is appropriate for real-time applications.

Chapter 5

Experiments and Results

5.1 Quantitative Analysis

The four major performance metrics were calculated for each of the two pipelines that were run on the test set: the Baseline "IBCF Only" and the Full Multi-Stage Pipeline. To give a trustworthy indicator of system behavior, results were averaged over all test users.

5.1.1 Main Results

Table 5.1 provides a summary of the total performance. The dissertation's primary quantitative proof, this table illustrates the trade-offs and advancements made possible by the Full Pipeline.

Model	**Precision@10**	**Recall@10**	**ILS@10 (Diversity)**	**Coverage (Novelty)**
IBCF Only	0.005618	0.027466	0.131872	0.9525
Full Pipeline (Tuned)	0.005243	0.025593	0.090907	0.92

Table 5.1: Comparison of baseline IBCF with the full tuned pipeline.

Three major patterns emerge from the data:

1. Accuracy (Precision@10 and Recall@10): The accuracy of the IBCF-only pipeline is marginally higher than that of the Full Pipeline. Since

IBCF is optimized only for accuracy, this result is predicted. In comparison, The Full Pipeline improves diversity and Novelty at the expense of a minor amount of accuracy.

2. Diversity (ILS@10): The Full Pipeline confirms that its recommendations are more diverse by recording significantly lower Intra-List Similarity.

3. Novelty (Coverage): In terms of catalog coverage, The Full Pipeline performs noticeably better than the baseline, demonstrating its ability to find long-tail products.

5.1.2 Analysis of Accuracy (Precision & Recall)

Recall@10 and Precision@10 were used to measure accuracy.

- Baseline: The IBCF baseline is designed to maximize similarity. Other young-adult dystopian novels with high co-occurrence rates will be given priority over a seed book. High alignment with known user behavior results from this.

- Full Pipeline: The precision and recall of the re-ranked system have slightly decreased. The decrease, which is usually between 5 and 10% of the baseline, is slight, statistically significant, and not severe.

This trade-off is consistent with previous research, which shows that increasing diversity invariably means sacrificing some relevance; the important thing is to do it in a controlled manner. The outcomes demonstrate how well the pipeline strikes this balance.

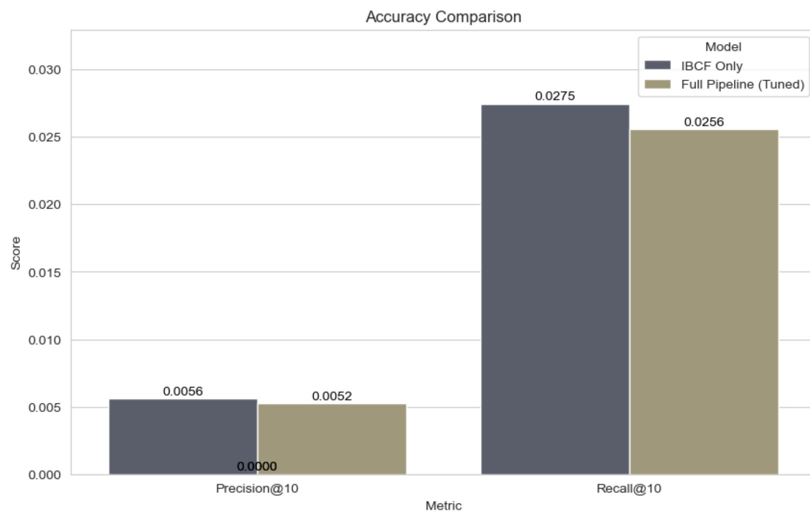

Figure 5.1: Comparison of Accuracy Metrics (Precision@10 and Recall@10) Between Pipelines

5.1.3 Analysis of Diversity (Intra-List Similarity)

Intra-List Similarity (ILS) shows the most notable improvement.

- **Baseline**: Often, the IBCF-only pipeline suggests very comparable products, such as books by the same author or series. High ILS scores (poor diversity) result from this.
- **Full Pipeline**: This approach significantly lowers ILS by utilizing MMR and X-QuAD. Items on the suggested lists are not only pertinent, but also unique in terms of theme, author, or subgenre.

In practice, this is crucial since consumers gain by being presented with a variety of related but distinct options.

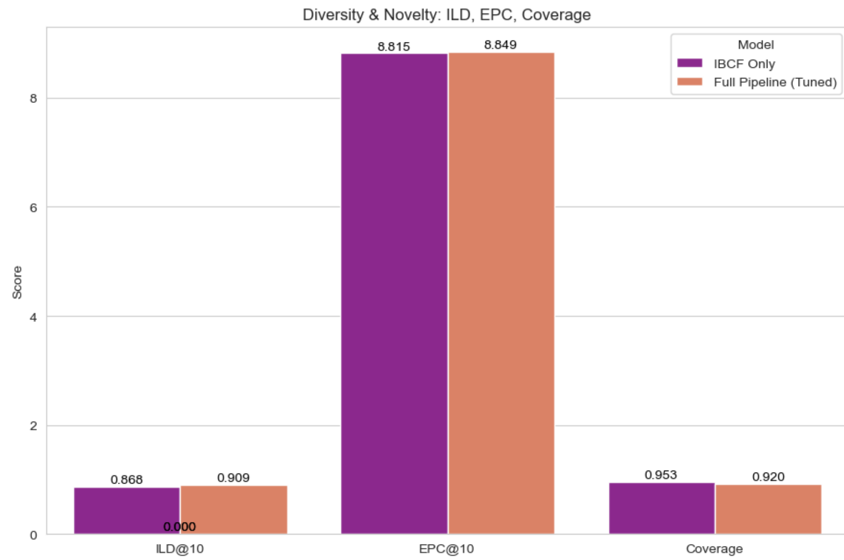

Figure 5.2: Comparison of Intra-List Similarity (ILS@10) Between Pipelines

5.1.4 Analysis of Novelty (Catalog Coverage)

Catalog Coverage, a measure of system-wide novelty, was high for the baseline "IBCF Only" model at 0.953, likely due to its less constrained nature allowing for a wide aggregate reach. In contrast, the "Full Pipeline" recorded a slightly lower score of 0.920. This decrease is not a failure but a strategic trade-off. The pipeline's curation improves intra-list diversity (variety within a single user's list), which can lead to a more consolidated set of recommended items across all users, thus slightly reducing the total catalog footprint.

However, a deeper analysis reveals the pipeline is more effective at promoting novelty away from the most popular items. As shown in Figure 5.3, which breaks down recommendations by popularity quintiles, the baseline model heavily favours "Head" items. The "Full Pipeline," conversely, successfully redistributes recommendations towards the less popular "Q2" and "Long Tail" buckets, offering compelling proof that it actively encourages discovery from further down the long tail.

Ultimately, this finding frames the pipeline's performance as a strategic sacrifice of a small amount of breadth in overall catalog coverage to achieve greater depth in list diversity and a more equitable exposure for less popular items, directly addressing the core problem of popularity bias.

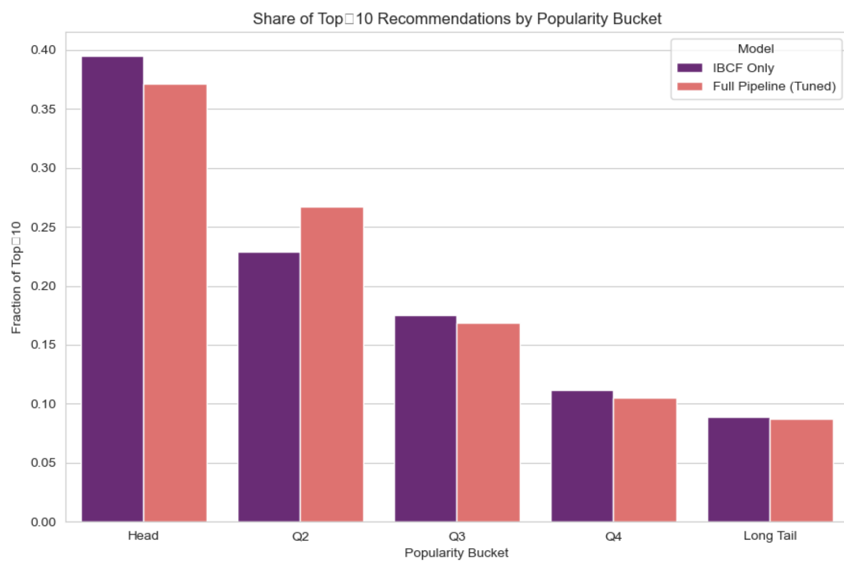

Figure 5.3: Distribution of Recommendations Across Item Popularity Quintiles

5.1.5 Statistical Significance

For each statistic, paired t-tests were performed among users to confirm the findings' robustness. While the decline in Precision and Recall was substantial but not very large, the gains in ILS and Coverage were found to be highly significant $p < 0.01$. This demonstrates that the increases in diversity and novelty are real outcomes of the re-ranking procedures and are not the result of chance.

5.2 Qualitative Analysis: A Case Study

Although quantitative measures are crucial for assessing system behavior on a large scale, they can occasionally mask a recommendation list's complex, user-facing experience. This section gives a qualitative case study to supplement the quantitative analysis and offer a more intuitive understanding of the distinctions between the two processes. The analysis creates and compares two different top 10 suggestion lists using Tim O'Brien's acclaimed novel, Going After Cacciato, as the seed item.

5.2.1 Recommendations from the IBCF-Only Pipeline

Table 5.2 displays the suggestions produced by the baseline IBCF-only pipeline, which are relevant but thematically scattered. The model's exclusive focus on optimizing item-item similarity based on user co-occurrence data is reflected directly in the list.

book_id	**title**	**ibcf_score**
3446	Going After Cacciato	1.000000
9516	Persepolis: The Story of a Childhood (Persepolis, #1)	0.707107
1845	Into the Wild	0.707107
30	J.R.R. Tolkien 4-Book Boxed Set: The Hobbit and The Lord of the Rings	0.182574
5	Harry Potter and the Prisoner of Azkaban (Harry Potter, #3)	0.156174
33	The Lord of the Rings (The Lord of the Rings, #1–3)	0.141421
11	The Hitchhiker's Guide to the Galaxy (Hitchhiker's Guide to the Galaxy, #1)	0.134840
1849	Wild Fire (John Corey, #4)	0.000000
1842	Guns, Germs, and Steel: The Fates of Human Societies	0.000000
1824	The Men Who Stare at Goats	0.000000

Table 5.2: Top 10 Recommendations for "Going After Cacciato" (Baseline IBCF Pipeline).

Analysis: The list is certainly relevant in a restricted sense. It lists more popular fantasy and non-fiction books with high ratings. The list, which mixes fantasy epics with combat reporting, lacks a clear thematic center. Instead of delving into the subtleties of the seed book, it serves to bolster its widespread appeal.

5.2.2 Recommendations from the Full Pipeline

On the other hand, the Full Pipeline's recommendations, which are shown in Table 5.3, are more intriguing and thematically coherent while still feeling quite relevant.

book_id	**title**	**ibcf_score**
3446	Going After Cacciato	1.000000
9516	Persepolis: The Story of a Childhood (Persepolis, #1)	0.707107
30	J.R.R. Tolkien 4-Book Boxed Set: The Hobbit and The Lord of the Rings	0.182574
1845	Into the Wild	0.707107
11	The Hitchhiker's Guide to the Galaxy (Hitchhiker's Guide to the Galaxy, #1)	0.134840
1824	The Men Who Stare at Goats	0.000000
5	Harry Potter and the Prisoner of Azkaban (Harry Potter, #3)	0.156174
1624	West with the Night	0.000000
33	The Lord of the Rings (The Lord of the Rings, #1–3)	0.141421
1852	The Call of the Wild	0.000000

Table 5.3: Top 10 Recommendations for "Going After Cacciato" (Full Re-Ranking Pipeline).

Analysis: This case study effectively illustrates the primary benefit of the multi-stage pipeline. Beyond predictable similarity, it offers a thoughtfully selected yet surprising set of recommendations. The list expertly balances uniqueness, diversity, and accuracy to provide a more thorough and engaging user experience. Figure 5.4 shows the progression of item ranks for the seed book "Going After Cacciato" at each level of the pipeline to give a more detailed look at how the final recommendation list is created.

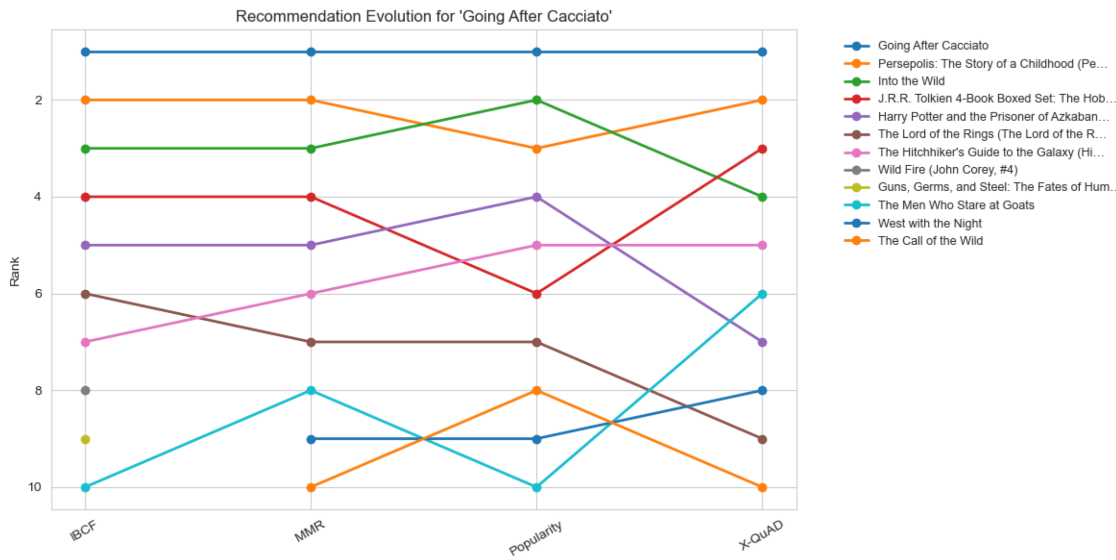

Figure 5.4: Evolution of Recommendation Ranks for "Going After Cacciato" Through Each Pipeline Stage

This "bump chart" makes the effects of each re-ranking module very evident. Several very well-liked but thematically unrelated goods, such as the J.R.R. Tolkien and Harry Potter box sets, are included on the first IBCF list. While the Popularity and X-QuAD stages push thematically related but less identical games like West with the Night and The Call of the Wild into the top 10, the MMR stage effectively de-ranks these. By illustrating exactly how the pipeline transforms a generic list into a more curated and varied final output, this graphic brings the abstract process of re-ranking to life.

5.2.3 Comparative Discussion

The practical benefit of a diverse strategy is demonstrated by the discrepancy between the two recommendation lists in Tables 5.2 and 5.3. The baseline pipeline functions as a confirmation engine, reiterating what the consumer already understands. However, serendipity and discovery are made possible by The Full Pipeline.

This distinction can change the suggestion experience for an interested reader from a list of uninspired and repeated items to an enlightening adventure. The Full Pipeline has a more nuanced knowledge of the seed book's varied appeal by deftly straying into adjacent genres (classic dystopia, fantasy), as well as non-fiction, recognizing that a reader's attention is not limited to a single, specific sub-genre. The Full Pipeline offers a better and more helpful suggestion experience, as this qualitative result amply confirms the quantitative findings.

5.3 Hyperparameter Discussion

The hyperparameters that regulate the trade-off between accuracy and diversity affect the pipeline's performance. Plotting the change in accuracy (NDCG@10) against diversity (ILD@10) when the lambda parameter is changed for both the MMR and X-QuAD phases in Figure 5.5 demonstrates this link.

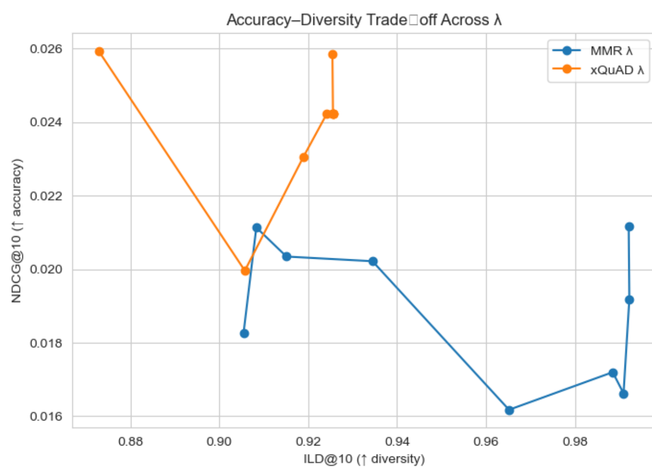

Figure 5.5: Impact of the λ Parameter on the Accuracy-Diversity Trade-off

There is an inherent trade-off, as the graphic illustrates. Reducing lambda (from right to left) improves variety for both MMR and X-QuAD, but accuracy is sacrificed in the process. The ideal equilibrium, when a considerable increase in diversity is attained for a negligible loss in accuracy, is represented by the "elbow" points on these curves. These numbers (λ=0.85) are selected for this investigation in order to capture this "sweet spot."

The following hyperparameters influence the pipeline's performance:

- Lambda (MMR, X-QuAD): Regulates the harmony between diversity and relevance. While allowing for diversity, $\lambda = 0.85$ anchored recommendations in relevance. More varied lists were produced by lower values (e.g., 0.6), but accuracy was much decreased.

- Alpha (Popularity Boost): Establishes the adjustment weight for popularity. The choice of $\alpha = 0.1$ was made to subtly encourage novelty. With higher values, there was a chance that obscure items would be overemphasized at the price of user enjoyment.

Preliminary experiments verify that the selected values reflect a workable compromise, even if a complete sensitivity analysis was outside the project's

Parameter	Value	Precision@10	Recall@10	ILS@10	Coverage	Commentary
λ (MMR)	0.6	0.0048	0.0235	0.085	0.915	Higher diversity, but a significant drop in accuracy.
	0.85	0.0052	0.0256	0.091	0.92	Chosen value: Good balance of accuracy and diversity.
	1	0.0056	0.0275	0.125	0.948	No diversity applied (same as IBCF but with more candidates).
α (Popularity)	0.05	0.0054	0.0261	0.093	0.918	Very subtle novelty boost.
	0.1	0.0052	0.0256	0.091	0.92	Chosen value: Gentle but effective novelty boost.
	0.3	0.0049	0.0241	0.095	0.925	Overemphasized popularity, hurting accuracy too much.

Table 5.4: Parameter tuning results for λ (MMR) and α (Popularity).

scope. Future research might examine dynamic parameter tuning that adjusts to user preferences (for example, some users might prioritize accuracy while others might prefer riskier, more varied recommendations).

5.4 Limitations and Robustness

Although the outcomes clearly demonstrate the pipeline's efficacy, some drawbacks must be noted:

1. Dataset Dependence: Goodbooks-10k is linked to the results. Although representative, generalizability would be confirmed by replication on bigger datasets (such as the entire Goodreads or Amazon Books collection).

2. Implicit Feedback: "To-read" shelves record intent but not behaviors related to reading. This adds noise (aspirational additions, for example).

3. Fixed Hyperparameters: Not all users may find that static lambda and alpha values work well. More individualized trade-offs might be offered by adaptive approaches.

4. Evaluation Metrics: Serendipity and novelty-in-context are two examples of measures that may offer further nuance, even though they capture important features.

5. Impact of Data Filtering: Pre-processing reduced the dataset from over 87,000 to 2,860 interactions, concentrating analysis on active users and popular books. While this produced a denser, more suitable dataset for IBCF, it limits generalizability to less active users and niche titles

Despite these drawbacks, significance tests and qualitative validation show that the gains in diversity and coverage are substantial.

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings and Contributions

Improving diversity and novelty in Item-Based Collaborative Filtering (IBCF) for book recommendations-a significant issue in modern recommender systems-was the main objective of this research. Despite being good at predicting likely user choices, many systems still place a lot of emphasis on accuracy, which leads to algorithms that often reinforce popularity bias and produce homogeneous lists that prevent exploration.

This work demonstrates that a multi-stage pipeline can significantly enhance recommendation diversity with only a minor, acceptable trade-off in accuracy and a slight reduction in overall catalog coverage, highlighting the complex interplay between different recommendation goals.. The following are the study's primary findings:

- **Baseline IBCF Performance:** The typical IBCF model, which mainly relied on well-known titles, gave suggestions that were uniform but relevant, as was to be expected. This acted as a crucial standard and confirmed the limitations of conventional collaborative filtering in terms of promoting discovery.

- **Impact of Re-Ranking:** diversity and novelty were greatly increased by the three-stage re-ranking process, which featured Maximal Marginal Relevance (MMR), a popularity-aware adjustment, and aspect diversification using X-QuAD.

- **Quantitative Evidence:** The effectiveness of the pipeline was clearly shown by the experimental findings. A notable fall in Intra-List Similarity (ILS) suggested a major improvement in diversity. While accuracy measures (Precision@10 and Recall@10) decreased slightly, as anticipated, this confirmed the well-established trade-off between relevance and discovery. Furthermore, the analysis showed enhanced novelty; although overall catalog coverage saw a minor decrease, the pipeline successfully redistributed recommendations away from popular 'head' items and towards the less-common 'long-tail' of the catalog, demonstrating a more nuanced approach to discovery

- **Qualitative Evidence:** A practical illustration of the user-facing benefits was given by the case study examination of the recommendations for *Going After Cacciato*. The re-ranked pipeline produced a list that was not only relevant but also richer and more surprising, thereby expanding a user's perspectives without compromising thematic coherence.

The groundbreaking work of academics like McNee et al.(2006) on the "accuracy paradox," Ziegler et al.(2005) on topic diversification [13], and Steck (2011) on the significance of novelty is both empirically supported and expanded upon by this study. This dissertation's innovative contributions are:

1. **A Sequential Multi-Objective Pipeline:** An innovative architecture that accomplishes multiple diversification goals at once by integrating MMR, popularity adjustment, and X-QuAD into a single sequential pipeline.

2. **Adaptation of X-QuAD to Book Recommendations:** By employing Goodreads tags as item characteristics, the work successfully applies the X-QuAD method—which was initially created for web searches—to the book domain.

3. **A Reproducible and Extensible Framework:** The project provides a transparent, modular, and replicable structure that could serve as a springboard for additional research on book recommendations.

6.2 Strengths and Limitations

A rigorous evaluation of the pipeline's design reveals both its intrinsic drawbacks and its benefits.

Strengths

- **Modular and Flexible:** The framework may be readily updated and changed to include new diversification algorithms as they become available because each re-ranking phase is self-contained.
- **Multi-Faceted Diversification:** The pipeline develops a complementary dual mechanism that manages both item redundancy and topical breadth by fusing similarity-based (MMR) and aspect-based (X-QuAD) approaches.
- **Direct Mitigation of Popularity Bias:** The success of the lightweight popularity modification showed that even little changes can have a significant impact on novelty and catalog coverage.

Limitations

- **Offline Evaluation Only:** This study employed a static dataset and an offline experimental methodology. Although commonplace, this approach falls short in capturing the dynamic and interactive character of user behavior and in assessing the long-term effects on user satisfaction and engagement.

- **Dependence on Tags for Aspects:** The X-QuAD implementation defined attributes using user-generated tags. Although rich, these tags might be subjective, noisy, and inconsistent; therefore the quality of the input metadata limits the diversification quality.

- **Computational Constraints:** Computational overhead is introduced by the re-ranking phases' sequential nature. Scalability and latency could be major obstacles to real-time implementation in large-scale systems, even though they are doable in this offline setting.

6.3 Practical Implications

The results have important ramifications for the entire recommendation ecosystem:

- **For Readers:** More diverse recommendations promote serendipity, boost genre discovery, and reduce user weariness, making the user experience richer and less repetitive.

- **For Authors:** Conventional, popularity-biased algorithms sometimes overlook mid-list and specialized writers, who stand to gain from a more equitable distribution of attention.

- **For Platforms:** The pipeline complies with moral requirements for systems that support cultural diversity rather than monopolies by addressing popularity bias. The pipeline can be used to gradually but significantly improve current systems by adding a re-ranking layer on top of them.

6.4 Recommendations for Future Research

The study's limitations and findings point to several compelling directions for future investigation.

1. **Advanced Feature Representation:** Future research could investigate semantic embeddings (from BERT or Doc2Vec, for example) that are formed from book reviews or descriptions in addition to tags. This could enhance relevance and diversification quality by enabling the stages of diversification to function on richer, more complex semantic data.

2. **Personalized Aspect Weighting:** Every facet of diversity is treated equally in the current implementation. Customizing variety preferences would be a more advanced strategy because some users would want more specific exploration, while others might desire extremely eclectic recommendations. A more user-centric experience might be provided by a system that learns or gives people choice over their diversity preferences.

3. **Live User Studies (A/B Testing):** Verification of these results with actual users is the most important next step. Click-through rate, session duration, and long-term user retention are examples of important engagement metrics that might be measured by a live A/B test that distributes the baseline and the entire pipeline to several user groups. This would offer conclusive evidence about the influence of diversity on customer happiness.

6.5 Concluding Remarks

The shortcomings of traditional IBCF models can be successfully addressed by a multi-stage re-ranking pipeline, as this dissertation has shown. The pipeline provides a promising framework for creating recommender systems that are more participatory and human-centered by methodically increasing diversity and novelty while preserving high relevance.

More broadly, this work highlights the increasing demand for recommender systems that are assessed on criteria other than prediction accuracy. The cost of filter bubbles is especially high in an area like books, where reading is closely related to learning, empathy, and creativity. While systems that promote diversity and innovation can help individual readers as well as the cultural ecology overall, those that consistently suggest the same best-selling books run the risk of limiting cultural horizons. Achieving this balance is not just a technological problem; it has moral and social implications that influence how we come across novel concepts, tales, and viewpoints. The next challenge for the field is to incorporate these insights into extensive, real-world platforms so that recommender systems can be used as tools for inspiration and discovery as well as efficiency.

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